Davashis Deb

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EDUCATION

Shahjalal University of Science and Technology (SUST) Dec 2019 - July 2025
Bachelor of Science in Civil Engineering (Sylhet Engineering College) Sylhet, Bangladesh
Cumulative GPA: **3.57 / 4.00**
Undergraduate Thesis: *Geotechnical applications of construction and demolition waste and future prospects in Bangladesh - a comprehensive review.* Supervised by Prof. Dr. Shriful Islam.
Key Focus: Evaluated the geotechnical properties of construction and demolition waste (CDW) and its effectiveness in soil stabilization and ground improvement. It mainly focused on the sustainable application of CDW in Bangladesh's infrastructure, to reduce environmental impact and provided a framework for using these waste materials as sustainable alternatives to depleting natural aggregates.

RESEARCH INTERESTS

- Geotechnical Engineering
- Ground Improvement Techniques
- Slope Stability
- Foundation Design
- Sustainable Construction Materials
- Soil Stabilization
- Environmental Geotechnics
- Sustainable Design of Base, Sub-base, and Sub-grade

RELEVANT COURSEWORK

CE 541	Geotechnical Engineering I	**CE 643**	Geotechnical Engineering II
CE 845	Geotechnical Engineering III	**CE 542**	Geotechnical Eng. Sessional I
CE 844	Geotechnical Eng. Sessional II	**CE 600**	Research Tools and Techniques
CE 785	Fundamental of Earthquake Eng.	**STA 201**	Statistics
CE 481	Eng. Geology and Geomorphology	**CE 309**	Numerical Analysis
CE 450	Remote Sensing and GIS	**CE 683**	Project Management

RESEARCH AND PROFESSIONAL EXPERIENCE

Shahjalal University of Science and Technology (SUST) Feb 2026 – Present
Graduate Research Assistant Sylhet, Bangladesh

- Researching sustainable soil stabilization techniques using alkali-activated Areca Husk and Incense Stick ashes.
- Simulating and validating the bearing capacity and settlement characteristics of stabilized soils using advanced finite element analysis.

Sylhet Engineering College Jan 2022– Jul 2025
Undergraduate Research Assistant Sylhet, Bangladesh

- Conducted a comprehensive review of Construction and Demolition Wastes (CDW) utilization in geotechnical applications.
- Explored prospects of CDW-based materials in the context of Bangladesh.
- Conducted a PLAXIS 3D finite-element analysis on spread footings to evaluate the effect of a macadam layer on allowable bearing capacity.
- Co-authored a waste-to-energy assessment for Sylhet City Corporation, estimating daily energy potential from municipal solid waste.

FAZ Engineering & Construction (Freelance) Jan 2024 – Present
3D BIM Modeler Sylhet, Bangladesh

- Created 3D BIM models for 15+ residential projects using Revit.
- Rendered interior and exterior views and videos using Enscape.
- Visited and surveyed several construction sites.

Design & Development Solution (DDS) April 2024 – June 2024
Intern – Basic RCC Structural Design Sylhet, Bangladesh

- Performed structural modeling of residential buildings using ETABS.
- Calculated loads acting on the foundation and designed structural components.
- Checked the serviceability and overall stability of the building model.

LANGUAGE PROFICIENCY

Bangla **-** Native
English - **C1 (IELTS Academic – Band Score: 7.0)**

PUBLICATIONS

Journal Articles

1. Shuvo, B. P., **Deb, D.**, Smrity, T. A., Islam, J., & Islam, S. (2026). Geotechnical applications of construction and demolition waste and future prospects in Bangladesh - a comprehensive review. *Discover Civil Engineering* (Springer), 3(1), 27. *https://doi.org/10.1007/s44290-026-00418-w*

Conference Papers

1. Hasan, M. T., Mahin, H. D., **Deb, D.**, & Shuvo, B. P. (2024). ANALYSIS OF SOIL BEARING CAPACITY PROVIDING MACADAM LAYER UNDER SPREAD FOOTING USING PLAXIS 3D. *7th International Conference on Civil Engineering for Sustainable Development (ICCESD 2024), KUET. https://bit.ly/4uzeHvC*
2. Dev Mahin, H., **Deb, D.**, Paul Shuvo, B., Titumir Hasan, M., & Atiqul Hasan, M. (2023). Waste To Energy Potential in Sylhet City Corporation Area. *7th International Conference on Engineering Research, Innovation and Education School of Applied Sciences & Technology (ICERIE, 2023), SUST. https://dx.doi.org/10.2139/ssrn.5136742*

TECHNICAL SKILLS

Programming: Python (Basic)
Document Creation: Microsoft Office
Simulation Software: Plaxis 2D/3D/LE
Engineering Software: ArcGIS, Revit, Enscape, ETABS, AutoCAD

CERTIFICATIONS

Certificate of Attendance: *Teacher Seminar on Japanese Language Education for Life in Japan,* Organized by The Japan Foundation and the Institute of Modern Languages, University of Dhaka (Sylhet, May 2026).

Certificate of Participation: *Truss Trial* at exCEEd 2023, organized by the Department of Civil and Environmental Engineering, SUST (March 2023).

Conference Attendance: *International Conference on Engineering Research, Innovation and Education (ICERIE 2023)*, Shahjalal University of Science and Technology (Jan 2023).

Certificate of Appreciation for Leadership: Awarded by the *Department of Civil Engineering, Sylhet Engineering College*, for valuable contributions and dynamic leadership as an Executive Member of the SEC Civil Society (July 2025).

REFERENCES

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